

Chandler Higgins

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EXPERIENCE

Software Development Intern

May 2026 – August 2026

Amazon Web Services

Vancouver, BC

- Engineered a distributed internal operations platform using Java, React, and TypeScript, consolidating 5 fragmented operational CLIs into a unified Object Explorer alongside 4 additional investigation tools, deployed and used by the Amazon SNS on-call rotation.
- Architected an asynchronous backend execution engine to securely execute host-level operational CLIs, parsing unstructured text outputs into strongly typed Java objects with robust error-boundary handling.
- Designed a dynamic workflow routing layer that parses the Amazon Resource Name (ARN) to automatically adapt investigative tools based on AWS SNS Standard vs. FIFO topic configurations.
- Implemented an enterprise-grade security workflow for SNS topic-policy recovery, enforcing POSIX-group authorization, strict authentication, and a mandatory two-person review (2PR) consensus protocol.

Software Development Intern

May 2025 – August 2025

Amazon Web Services

Vancouver, BC

- Developed and deployed an extensible host-level Health Check engine critical to maintaining global service reliability for AWS SQS and SNS at hyperscale.
- Authored cross-platform diagnostic checks executing directly on Amazon Linux hosts to monitor RPM database consistency and Graviton CPU hardware functionality.
- Built a configurable telemetry and metrics system using Spring, exposing real-time visibility into regional deployment status, server health, and failure modalities.
- Utilized Spring to convert object-oriented methods into executable host scripts, automating deployment workflows and integrating seamlessly with existing infrastructure platforms.

PROJECTS

DialPass: AI Phone-Tree Agent | *Python, FastAPI, Twilio, AWS SQS, Docker*

August 2026 – Present

- Built a bidirectional audio engine with FastAPI WebSockets and `asyncio` that streams live Twilio call audio to OpenAI's Realtime API, letting an AI agent navigate phone trees and hand a live human to the user.
- Designed a two-tier inference state machine that classifies audio cheaply during hold and invokes the full real-time model only on menu prompts or agent pickup, reducing active AI inference by up to 90%.
- Architected an AWS SQS telemetry pipeline to keep call-event data off the critical audio path, with a circuit breaker that reroutes dropped calls and timeouts to a fallback notification.

Custom Dynamic Music Analytics Engine | *Java, Spring Boot, PostgreSQL, Docker*

June 2026

- Engineered a declarative music analytics platform allowing users to design and execute custom aggregation queries over unstructured multi-year streaming history datasets.
- Designed a dynamic query compilation layer in Java to parse user-defined statistical metrics into highly optimized, multi-stage PostgreSQL processing pipelines.
- Optimized analytical performance by implementing database indices, caching layers, and parallel data streams to reduce execution latency across million-row JSON listener logs.

TECHNICAL SKILLS

Languages: Java, TypeScript, Python, Kotlin, C, JavaScript, SQL, HTML/CSS

Frameworks and Technologies: FastAPI, AWS SQS/SNS, Spring Boot, PostgreSQL, React, Docker, Twilio API, JUnit, Mockito, Git

EDUCATION

Concordia University

Montreal, QC

Bachelor of Computer Science

Expected December 2027

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Theory of Computation, Artificial Intelligence